

Lecture 3

07/08/24 10:02 AM

Continuity: $LHL = RHL = f(x)$

For a continuous fn $f(\cdot)$, as $x \rightarrow c$
we have $f(x) \rightarrow f(c)$.

Consider a monotonic sequence of sets $A_1, A_2, \dots, A_n$.

For a continuous set fn S , as $A_n \rightarrow A$,
we have $S(A_n) \rightarrow S(A)$. (limiting set)

$\mathbb{P}$ is a continuous set function.

• SEQUENCE OF SETS:

Given $(\Omega, \mathcal{F})$, if $A_1 \subset A_2, \dots$ is an increasing sequence of events defined on $\mathcal{F}$ and $\bigcup_{n=1}^{\infty} A_n = A \in \mathcal{F}$, then we say the sequence of sets A_n is increasing to A . ($A_n \uparrow A$) lim. A_n for $\bigcup_{n=1}^{\infty} A_n$

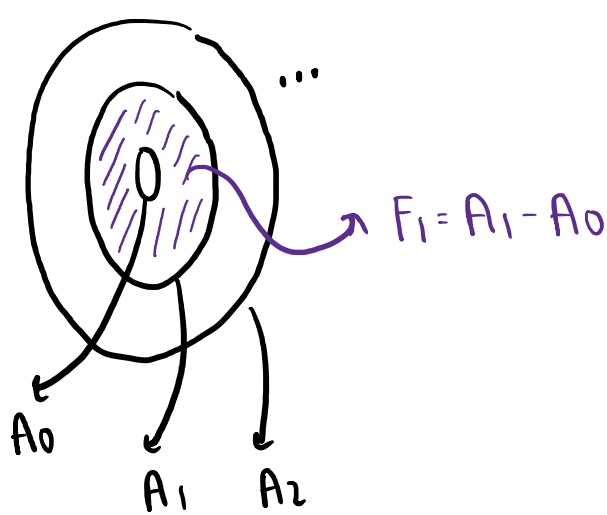
Similarly, $A_1 \supset A_2, \dots$, $\bigcap_{n=1}^{\infty} A_n = A$ is a decreasing sequence of events. ($A_n \downarrow A$) lim. A_n for $\bigcap_{n=1}^{\infty} A_n$

LEMMA:

For sequence of events of the type $A_n \uparrow A$ or $A_n \downarrow A$
 $\lim_{n \rightarrow \infty} \mathbb{P}(A_n) = \mathbb{P}(A)$

$\bigsqcup_{i=1}^{\infty} F_i$
disjoint union

$$\mathbb{P}\left(\bigsqcup_{i=1}^{\infty} F_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(F_i)$$



$$F_n = A_n - A_{n-1}$$

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} F_n$$

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} F_i\right)$$

$$\lim_{n \rightarrow \infty} \mathbb{P}(A_n) = \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{i=1}^n F_i\right) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(F_i) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$$

↓
disjoint union

CONDITIONAL PROBABILITY:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}, \quad \mathbb{P}(A) > 0$$

$$\mathbb{P}(A|B) \cdot \mathbb{P}(B) = \mathbb{P}(B|A) \cdot \mathbb{P}(A)$$

$$\text{BAYE'S RULE: } \mathbb{P}(B|A) = \frac{\mathbb{P}(A|B) \cdot \mathbb{P}(B)}{\mathbb{P}(A)}$$

$$\mathbb{P}(A|(B \cap C)) = \mathbb{P}(A|BC)$$

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A) \cdot \mathbb{P}(B|A) \cdot \mathbb{P}(C|AB)$$

CHAIN RULE

- Consider a finite sample space Ω where each outcome is equally likely.
Then, $\mathbb{P}(B|A) = \frac{|A \cap B|}{|B|}$

LAW OF TOTAL PROBABILITY:

Consider a sample space Ω divided into disjoint subsets $B_1, B_2, \dots, B_n$.

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}\left(\bigcup_{i=1}^n A \cap B_i\right) = \sum_{i=1}^n \frac{\mathbb{P}(A \cap B_i)}{\mathbb{P}(B_i)} \times \mathbb{P}(B_i) \\ &= \sum_{i=1}^n \mathbb{P}(A|B_i) \cdot \mathbb{P}(B_i) \\ &= \mathbb{P}(B_1) \cdot \mathbb{P}(A|B_1) + \mathbb{P}(B_2) \cdot \mathbb{P}(A|B_2) + \dots \end{aligned}$$

$$\mathbb{P}(A) = \mathbb{P}(A|B) \cdot \mathbb{P}(B) + \mathbb{P}(A|B^c) \cdot \mathbb{P}(B^c)$$

- Q. I have 3 bags that contain M marbles.
Bag i has R_i red marbles and B_i blue marbles.
I choose a bag at random and then draw a marble.
What is the probability that the chosen marble is red?

$$\text{ans. } \frac{1}{3} \times \frac{R_1}{R_1+B_1} + \frac{1}{3} \times \frac{R_2}{R_2+B_2} + \frac{1}{3} \times \frac{R_3}{R_3+B_3}$$

INDEPENDENCE

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B)$$

$$\begin{aligned} \mathbb{P}(A|B) &= \mathbb{P}(A) & \mathbb{P}(A|B^c) &= \mathbb{P}(A) \\ \mathbb{P}(B|A) &= \mathbb{P}(B) \end{aligned}$$

mutual independence $\Rightarrow$ pairwise independence
pairwise independence $\nRightarrow$ mutual independence

CORRELATION BETWEEN EVENTS

positively correlated: $\mathbb{P}(A|B) > \mathbb{P}(A)$
negatively correlated: $\mathbb{P}(A|B) < \mathbb{P}(A)$